

Ideation Phase

Smart Sorting & Idea Prioritization Template

Date	13 February 2026
Team ID	LTVIP2026TMIDS79181
Project Name	Smart Sorting – Rotten Fruit Detection using Machine Learning
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Problem Statement Selection

Problem Statement:

In the food supply chain and agricultural markets, sorting fruits manually is time-consuming and often inaccurate. Rotten or spoiled fruits mixed with fresh fruits can lead to quality degradation, food waste, and financial losses for vendors and distributors. Therefore, there is a need for an automated system that can accurately detect and separate rotten fruits from healthy ones using machine learning and image processing techniques.

Step-2: Brainstorm, Idea Listing and Grouping

1. Collect and prepare a dataset of fresh and rotten fruit images.
2. Preprocess images using resizing, normalization, and augmentation techniques.
3. Train a machine learning/deep learning model to classify fruits as fresh or rotten.
4. Implement a convolutional neural network (CNN) for image classification.
5. Evaluate model performance using accuracy, precision, recall, and loss metrics.
6. Develop a user interface to upload fruit images and display classification results.

Step-3: Idea Prioritization

High Priority:

- Dataset collection and preprocessing of fruit images
- Training a deep learning model for rotten fruit detection
- Evaluating model performance using accuracy and loss metrics

Medium Priority:

- Image augmentation and model optimization
- Developing a simple web interface for prediction

Low Priority (Future Scope):

- Real-time fruit detection using camera input
- Integration with automated fruit sorting machines
- Deployment of the model for large-scale industrial use